Serk Materials

Building Materials ERP and POS

Step-by-Step Testing Manual

Product	Serk Materials - cement, steel and hardware yard ERP and POS (offline-first, cloud sync)
Release under test	Release candidate for owner review - NOT production-approved
Stack	Django 5.2, Django REST Framework, SQLite (dev only) or PostgreSQL 16, ReportLab, openpyxl
Prepared	October 2026
Audience	Testers, accountants, yard and counter staff, owners performing acceptance, release engineers
Basis	README, ARCHITECTURE, DECISION_REGISTER (D01-D17), PRODUCT_SCOPE, RELEASE_READINESS, DEPLOYMENT, MIGRATION, BACKUP/RESTORE and OPERATIONS runbooks

Important. Screen labels in this manual (for example 'Cash desk - Open session') are taken from the project documentation. If a label on your screen differs slightly, record the actual label in the Notes box; that is a documentation finding, not necessarily a defect. Use **fictional data only** on any machine started with **seed_demo**.

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1 How to use this manual

1.1 Purpose

This manual tells a tester exactly what to click, what to type and what must happen, for every workflow the release candidate claims to support. It also covers synchronisation, backup and restore, deployment checks and the acceptance gates that the release notes list as 'NOT RUN'. Automated tests exist (section 17); this manual is the **human acceptance layer** on top of them and is the evidence the owners need to sign decisions D01-D17.

1.2 Test case format

Every test case is a box with an ID, a role, numbered steps, the expected result and a result column. Tick exactly one of Pass / Fail / Blocked. If Fail, write a defect number (Appendix A) in the Notes line. A test passes only if **every** bullet in the expected result is true.

Result	Meaning
Pass	All expected results observed exactly.
Fail	Any expected result not observed, an unexpected error page, or a wrong figure. Raise a defect.
Blocked	Could not run (missing data, environment, access). State why; re-run later.

1.3 Defect severity

Severity	Definition	Examples
Critical	Money, stock or security integrity wrong; data lost	Goods leave unpaid for a walk-in; stock does not reconcile; cashier sees cost; forged device writes stock
High	Core workflow blocked or wrong, no workaround	Cannot confirm an invoice; payment not settling invoice; sync never drains
Medium	Workflow works with a workaround or a wrong label/format	Wrong column in export; unclear error message
Low	Cosmetic	Alignment, spelling

1.4 Ground rules for testers

- Record the **starting figures** (stock, average cost, customer balance) before each money or stock test, and the ending figures after it. Most expected results are differences against the start.
- Never test on a live yard. Use a demo or training database, or a restored copy of production.
- Posted documents cannot be edited or deleted by design. A mistake is corrected with a void, credit note or adjustment - so plan the order of your tests and note document numbers as you go.
- Keep a running list of document numbers you create (PRO, INV, DO, RCT, GRN, TRF, ADJ, CRN) in the Test Data Sheet (Appendix C).
- Do not skip the negative tests (things that must be **refused**). They are the controls the owners are accepting.

2 Test environment and preparation

2.1 Environments

Environment	Used for	Profile / database
A. Single-machine demo	Sections 3-13 (functional tests)	SERK_PROFILE=dev, SQLite, seed_demo data
B. Cloud + edge pair	Section 14 (sync, devices, lease)	Cloud: PostgreSQL, SERK_NODE_ROLE=cloud. Edge: SERK_NODE_ROLE=edge, SERK_SITE_CODE=MKL1
C. Production-profile rehearsal host	Sections 15-16 (operations, deployment)	SERK_PROFILE=branch or cloud, PostgreSQL 16, TLS, systemd

2.2 Set up environment A (demo)

Run these commands once on the test machine:

```
python -m venv .venv
. .venv/bin/activate
pip install --no-deps --require-hashes -r requirements.lock
python manage.py migrate
python manage.py seed_demo
python manage.py runserver
```

Open **http://127.0.0.1:8000** in a browser (Chromium recommended; the system is tested on Chromium 141).

2.3 Test users (demo data only)

Username	Role	Password	Discount limit	Key capabilities
admin	Admin	serk12345	100%	Everything, including rebuild projections and maintenance
manager	Sales manager	serk12345	15%	Override credit, backorder, void, credit note, prices, verify payments, reports, cost and margin
cashier	Cashier	serk12345	3%	Sell, collect payment, open/close own cash session. No cost, no reports
yard	Storekeeper	serk12345	none	Dispatch, receive stock, transfer, adjust/count. Cannot sell or take payment
accounts	Accountant	serk12345	none	Collect and verify payment, void, credit note, reports, cost and margin

2.4 Baseline data to record before testing

Open the product list and customer list and fill in Appendix C. You need at least:

- **Product C** - a cement product (base unit KG; 1 BAG = 50 KG, 1 QTL = 100 KG, 1 TON = 1000 KG). Note on-hand quantity and average cost.
- **Product R** - a steel product sold by weight but stocked in pieces (for example rebar 12 mm: base PCS, 1 TON = 94 PCS).
- **Customer W** - the WALKIN cash customer.
- **Customer K** - a credit customer with a credit limit above zero and an available branch credit allowance. Note limit, allowance and current balance.
- **Customer Z** - a customer with credit limit 0 (treated as cash customer).
- Two yards (Yard 1 and Yard 2) for the transfer tests. If the demo has only one, ask admin to create the second before section 10.

If the demo data does not contain something above, create it as admin in *Administration* first and write it down. Do not invent figures in expected results: always compute them from your recorded start values.

3 Login, roles and access control

Before you start: Environment A running. Browser private window for each user so sessions do not mix.

AUTH-01 Every role can log in Role: all

Steps	Expected result	Result
1. Open the login page. **2.** Log in as admin / serk12345, note the menu, log out. **3.** Repeat for manager, cashier, yard and accounts.	- Each user reaches the home screen. - Menu entries differ by role (cashier has no Reports; yard has Dispatch and stock screens).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

AUTH-02 Wrong password refused Role: any

Steps	Expected result	Result
1. Enter username cashier with password wrong123. **2.** Enter an unknown username.	- Login is refused with a generic error message. - No server error page. Nothing reveals whether the user exists.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

AUTH-03 Cashier cannot see cost, margin or reports Role: cashier

Steps	Expected result	Result
1. Log in as cashier. Open the product list, a product, an invoice. **2.** Look for cost, margin or stock value columns. **3.** Type the reports address (`/reports/`) in the address bar.	- No cost, margin or stock value anywhere (screens, PDF, export). - Reports address is refused (403 or redirect), not displayed.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

AUTH-04 Storekeeper cannot sell or take money Role: yard

Steps	Expected result	Result
1. Log in as yard. **2.** Look for New sale and payment actions. **3.** Type the New sale address directly.	- No New sale or Collect payment button. - Direct address returns 403 (not authorised).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

AUTH-05 Accountant cannot dispatch or sell Role: accounts

Steps	Expected result	Result
1. Log in as accounts. **2.** Open an invoice. **3.** Look for Dispatch and Confirm actions.	- Dispatch and Confirm are absent or refused. - Collect payment, Void, Credit note, Reports and cost are available.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

AUTH-06 Manager cannot post stock Role: manager

Steps	Expected result	Result
1. Log in as manager. **2.** Look for Receive goods, Transfer, Adjust / count.	- These actions are absent or refused (manager has no receive/adjust capability). - Prices, void, credit override, verify payment are available.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

AUTH-07 Admin has full access Role: admin

Steps	Expected result	Result
1. Log in as admin. **2.** Open Administration, sales, stock, reports, Cloud sync.	- All areas open. - Posted documents in Administration are read-only (no edit/delete of invoices, payments, ledger).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

AUTH-08 Logout Role: any

Steps	Expected result	Result
1. Log out. **2.** Press the browser Back button.	- Session ended; Back does not show protected data without a new login.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

4 Catalog, units and prices

Before you start: Product C with unit conversions and at least two price lists (retail, wholesale or contractor).

CAT-01 Unit conversion is applied Role: cashier

Steps	Expected result	Result
1. Start a New sale. Add Product C in BAG, quantity 2. **2.** Change the unit to QTL (quantity 1) on a second line. **3.** Check the base quantity shown or on the confirmed invoice.	- 2 BAG = 100 KG; 1 QTL = 100 KG; 1 TON would be 1000 KG. - Each line shows its unit and price per that unit.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

CAT-02 Price list follows the customer Role: cashier

Steps	Expected result	Result
1. New sale for Customer W (retail). **2.** Note unit price of Product C. **3.** New sale for a wholesale/contractor customer. Note the price.	- Prices differ per price list as configured. - The customer panel shows price list, balance and available branch credit.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

CAT-03 Price change is effective-dated and audited Role: manager

Steps	Expected result	Result
1. Products - open Product C prices. Note current BAG price P0. **2.** Enter new price P1 and use Save new prices. **3.** Open a previously confirmed invoice for Product C. **4.** Create a new sale for Product C. **5.** Open the audit trail.	- Old confirmed invoice still shows P0. - New sale shows P1. - Audit trail has a PRICE_CHANGE entry with user and time.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

CAT-04 Cashier cannot change prices Role: cashier

Steps	Expected result	Result
1. Log in as cashier and look for Save new prices. **2.** Try the price-edit address directly.	- Not available / 403.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

CAT-05 Price derived for a unit with no explicit price Role: manager

Steps	Expected result	Result
1. Choose a unit for Product C that has no price row (for example TON). **2.** Add it to a sale.	- Price equals base-unit price multiplied by the factor (TON = 1000 x KG price).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

5 Cash sessions

Before you start: Cashier has no open session. Record the float to be used (for example 1,000.00).

CASH-01 Cash cannot be taken without an open session Role: cashier

Steps	Expected result	Result
1. Create and confirm an invoice for Customer W. **2.** Try to record a CASH payment.	- Payment is refused with a clear message that an open cash session is required.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

CASH-02 Open a session with a float Role: cashier

Steps	Expected result	Result
1. Cash desk - Open session. **2.** Enter the counted float (1,000.00). **3.** Confirm.	- Session is open and shows the float. - Only one open session per cashier.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

CASH-03 Close session without variance Role: cashier

Steps	Expected result	Result
1. Take two cash receipts (see PAY-01). **2.** Cash desk - Close session. **3.** Enter counted drawer = float + cash received exactly.	- Session closes. Variance 0.00. - No manager approval required.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

CASH-04 Variance needs manager approval Role: cashier, manager

Steps	Expected result	Result
1. Open a new session, take one cash receipt. **2.** Close with a counted amount that is 10.00 short. **3.** When asked, enter the manager's credentials.	- Close is refused without manager approval. - With manager credentials it closes; the variance and approver are recorded and audited.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

CASH-05 Daily cash report matches the session Role: manager

Steps	Expected result	Result
1. Reports - daily cash for today.	- Cash by method matches the sessions you closed. - Verified and pending amounts are shown separately.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

6 Sales: proforma, invoice, discounts, stock promise, credit

Before you start: Cashier session open. Note on-hand and available-to-promise (ATP) for Product C. ATP = on hand minus active reservations.

SAL-01 Create a proforma Role: cashier

Steps	Expected result	Result
1. New sale. Search for Product C and select it. **2.** Check it lands in the cart in its usual selling unit. **3.** Pick Customer K. **4.** Save as proforma.	- Customer panel shows price list, balance and available branch credit. - Proforma number follows SITE-PRO-YEAR-NNNNNN with no gaps. - Status PROFORMA, valid-until date set.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-02 Confirm invoice reserves stock and freezes prices Role: cashier

Steps	Expected result	Result
1. Open the proforma from SAL-01 and note ATP. **2.** Confirm invoice. **3.** Check ATP again. **4.** Change the product price (manager) and re-open the invoice.	- Invoice number SITE-INV-YEAR-NNNNNN assigned; status CONFIRMED. - ATP drops by the invoice quantity in base units; on hand unchanged until dispatch. - Invoice prices do not change after a price change.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-03 Discount within the cashier limit Role: cashier

Steps	Expected result	Result
1. New sale, add Product C. **2.** Apply a 3% discount to the line. **3.** Confirm.	- Accepted; line total reflects 3% off; audit has a DISCOUNT entry.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-04 Discount ceilings by role (3% / 15% / 100%) Role: cashier, manager, admin

Steps	Expected result	Result
1. As cashier apply 3.5%. **2.** As manager apply 15%, then 16%. **3.** As admin apply 50%.	- Cashier 3.5% refused. - Manager 15% accepted, 16% refused. - Admin accepted (limit 100%). - Each accepted discount is audited.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-05 Expired proforma cannot be confirmed Role: cashier, admin

Steps	Expected result	Result
1. Find or create a proforma whose valid-until date has passed. **2.** Run `python manage.py expire_proformas` (nightly job). **3.** Try to confirm it.	- Status becomes EXPIRED. - Confirmation is refused. A new proforma must be made.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-06 Cannot sell more than available Role: cashier

Steps	Expected result	Result
1. Note ATP of Product C (say N base units). **2.** Create an invoice for more than N (convert to base units). **3.** Confirm.	- Confirmation refused; message states stock is insufficient. - No invoice number consumed by a failed confirmation; no reservation created.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-07 Two confirmations cannot promise the same stock Role: cashier

Steps	Expected result	Result
1. Take a product with ATP of exactly 100 KG (use a GRN/adjustment to set it up). **2.** Create invoice A for 80 KG and invoice B for 80 KG (both proforma). **3.** Confirm A, then B.	- A confirms. B is refused (only 20 KG left to promise).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-08 Backorder only with the capability Role: cashier, manager

Steps	Expected result	Result
1. As cashier, try to confirm more than ATP (repeat SAL-06). **2.** As manager, confirm the same invoice declaring a backorder.	- Cashier refused. - Manager can confirm; the shortfall appears as backorder quantity on the invoice line and is visible to the customer on the printout.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-09 Credit within the branch allowance Role: cashier

Steps	Expected result	Result
1. Customer K: note limit, branch allowance and open balance. **2.** Confirm an invoice whose total is inside the available allowance, with no payment.	- Confirms. Customer balance rises by the total. - Available credit shown falls accordingly.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-10 Credit beyond allowance needs a manager Role: cashier, manager

Steps	Expected result	Result
1. As cashier confirm an unpaid invoice larger than available credit for Customer K. **2.** As manager repeat and use Override credit.	- Cashier refused. - Manager override succeeds; audit has a CREDIT_OVERRIDE entry with the approver.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-11 Customer with no credit must pay first Role: cashier

Steps	Expected result	Result
1. Confirm an invoice for Customer W (WALKIN) and another for Customer Z (credit limit 0), both unpaid.	- Confirmation may succeed, but both are treated as cash sales: dispatch will be blocked until paid (see GATE-01).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-12 Void an invoice before dispatch Role: manager

Steps	Expected result	Result
1. Confirm an invoice for Product C and note ATP. **2.** Void it with a reason. **3.** Try to void an invoice that already has a delivery.	- Status VOID; invoice number is kept (never reused); reservation released and ATP restored. - VOID_INVOICE audited. - Void after dispatch is refused (use a credit note).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-13 Proforma and invoice PDFs Role: cashier

Steps	Expected result	Result
1. Open a proforma and an invoice and download the PDF.	- PDF opens; correct number, customer, lines, VAT, totals. - Invoice PDF shows paid and balance.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SAL-14 Double-click on Confirm creates one invoice Role: cashier

Steps	Expected result	Result
1. Prepare a proforma. **2.** Double-click Confirm invoice rapidly (or press Back and re-submit).	- Exactly one invoice number is issued; no duplicate reservation.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

7 Payments and verification

Before you start: Cash session open for cashier. Two confirmed invoices ready: Invoice 1 for Customer W, Invoice 2 for Customer K.

PAY-01 Cash receipt is verified at once Role: cashier

Steps	Expected result	Result
1. Open Invoice 1, Collect payment, method CASH, full amount.	- Receipt number RCT issued; state VERIFIED immediately. - Invoice shows paid, balance 0.00; receipt belongs to the open session.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

PAY-02 Non-cash payment is pending Role: cashier

Steps	Expected result	Result
1. On Invoice 2 record a TELEBIRR payment with a reference (for example TB-TEST-001).	- Payment state PENDING. - Invoice is NOT settled; balance unchanged; pending amount visible on the daily cash report.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

PAY-03 Provider reference is mandatory Role: cashier

Steps	Expected result	Result
1. Record a BANK or CBE BIRR payment leaving the reference empty.	- Refused: reference required for non-cash.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

PAY-04 Duplicate provider reference refused Role: cashier

Steps	Expected result	Result
1. Record another payment (any invoice) with the same reference TB-TEST-001.	- Refused as duplicate; no second receipt.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

PAY-05 Second user verifies (segregation of duties) Role: cashier, manager/accounts

Steps	Expected result	Result
1. As the user who recorded it, open Cash desk - Payments awaiting verification and try to verify TB-TEST-001. **2.** Log in as manager (or accounts) and verify it against the statement.	- Recorder cannot verify their own payment. - A different user with verify capability can; state becomes VERIFIED; invoice settles; audit entry written.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

PAY-06 Reject a pending payment Role: manager

Steps	Expected result	Result
1. Record a CHEQUE with reference CHQ-001 as cashier. **2.** As manager reject/fail it.	- State FAILED; it never counts toward settlement; balance unchanged.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

PAY-07 Invalid amounts are rejected cleanly Role: cashier

Steps	Expected result	Result
1. Try amounts: 0, -50, abc, NaN, 1e999, empty.	- Every one is refused with a validation message (HTTP 400 style), never a server error page.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

PAY-08 Deposit on account Role: cashier/accounts

Steps	Expected result	Result
1. Record a payment for Customer K without selecting an invoice (deposit).	- Deposit is held unapplied; it appears in the receivables aging as unapplied deposit and reduces net exposure.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

PAY-09 Void a receipt Role: accounts/manager

Steps	Expected result	Result
1. Void the cash receipt from PAY-01 with a reason.	- Receipt voided; invoice balance restored; VOID_RECEIPT audited. - The voided receipt is NOT usable as refund evidence (see RET-05).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

PAY-10 Storekeeper cannot take payment Role: yard

Steps	Expected result	Result
1. Log in as yard and try to collect payment on an invoice.	- Not offered / refused (403).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

8 Gate: dispatch and release rule

Before you start: Storekeeper (yard) logged in on the gate screen (tablet-sized window is fine). Invoices from sections 6-7 available. Record stock of Product C and its average cost before each test.

GATE-01 Walk-in unpaid cannot leave Role: yard

Steps	Expected result	Result
1. Open an unpaid confirmed invoice for WALKIN. **2.** Dispatch.	- Dispatch refused: not settled and not on authorised credit. - No DO number issued; stock unchanged.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

GATE-02 Walk-in paid in full can leave Role: cashier, yard

Steps	Expected result	Result
1. Cashier takes CASH payment equal to the total. **2.** Yard dispatches the invoice.	- Dispatch accepted; DO number issued; status FULFILLED. - Stock falls by quantity x factor; movement is at the yard's average cost frozen at that moment.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

GATE-03 Pending payment does not release goods Role: cashier, manager, yard

Steps	Expected result	Result
1. Record a BANK payment (pending) for a walk-in invoice. **2.** Try to dispatch. **3.** Manager verifies the payment. **4.** Dispatch again.	- First dispatch refused. - After verification dispatch succeeds.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

GATE-04 Credit customer released on authorised credit Role: yard

Steps	Expected result	Result
1. Open an unpaid confirmed invoice for Customer K confirmed within the allowance (SAL-09). **2.** Dispatch.	- Dispatch allowed. Customer balance remains outstanding.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

GATE-05 Partial deliveries keep the remaining obligation Role: yard

Steps	Expected result	Result
1. Confirm an invoice for 100 BAG of Product C (pay or use credit). **2.** Dispatch 40 BAG. **3.** Open the invoice. **4.** Dispatch the remaining 60 BAG.	- After the first DO: status PARTIAL, delivered 40, remaining 60; stock down by 2,000 KG. - After the second DO: FULFILLED; total stock down 5,000 KG. - Each DO has its own number.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

GATE-06 Cannot dispatch more than remaining Role: yard		
Steps	**Expected result**	**Result**
1. On a PARTIAL invoice try to dispatch more than the remaining quantity.	- Refused with a clear message; no stock movement.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

GATE-07 Dispatch retry is safe Role: yard		
Steps	**Expected result**	**Result**
1. Double-click Dispatch, or submit, refresh and resubmit the same form.	- Only one delivery order and one stock deduction result.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

GATE-08 Delivery order PDF Role: yard		
Steps	**Expected result**	**Result**
1. Open the DO and download the PDF.	- Shows this delivery, base-unit quantity, ordered quantity and still-to-deliver; driver and plate if entered.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

GATE-09 Cashier cannot dispatch Role: cashier		
Steps	**Expected result**	**Result**
1. Log in as cashier and look for or try the Dispatch address.	- Not available / 403.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

GATE-10 Stock reconciles after dispatches Role: yard, manager		
Steps	**Expected result**	**Result**
1. Compare stock of Product C now with your starting figure minus every dispatch, in base units.	- Figures agree to 4 decimals. Average cost unchanged by sales.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

9 Returns, credit notes and refunds

Before you start: A fulfilled invoice for Product C exists. To prove 'original issue cost', first change the average cost with a GRN at a different cost (INV-01), then record the average cost before the return.

RET-01 Return and restock at the original issue cost Role: manager

Steps	Expected result	Result
1. Open the fulfilled invoice, Return goods. **2.** Return 5 BAG, disposition RESTOCK. **3.** Check stock, average cost and the credit note.	- Credit note number CRN issued with net and VAT split. - Stock rises by 250 KG at the ORIGINAL delivery issue cost, not the current average. - CREDIT_NOTE audited.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

RET-02 Damage and quarantine dispositions Role: manager

Steps	Expected result	Result
1. Return 2 BAG with disposition DAMAGE, and 2 BAG with QUARANTINE.	- Credit note recorded; goods are not restocked as normal saleable stock; the ledger shows the disposition.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

RET-03 Repeated partial returns reconcile to the cent Role: manager

Steps	Expected result	Result
1. On one invoice line (use an odd price such as 333.33) return in three partial returns that add up to the full line quantity.	- Sum of credit notes equals the invoiced line to the cent (final return absorbs the rounding).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

RET-04 Cannot return more than delivered Role: manager

Steps	Expected result	Result
1. Try to return more than the delivered quantity.	- Refused.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

RET-05 Refund is an explicit money movement Role: manager, cashier

Steps	Expected result	Result
1. On the credit note use Refund; manager authorises. **2.** Choose CASH from an open session (or a non-cash method with reference).	- Refund row records method, reference, session and credit note. - A cash refund reduces the open session's expected cash.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

RET-06 Cashier cannot issue credit notes Role: cashier		
Steps	**Expected result**	**Result**
1. Look for Return goods as cashier.	- Not available / 403.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

10 Inventory: receive, transfer, adjust, rebuild

Before you start: Storekeeper (yard) logged in. Record Product C stock quantity Q0 and average cost A0 in each yard.

INV-01 Goods receipt and weighted average cost Role: yard

Steps	Expected result	Result
1. Receive goods: 200 BAG of Product C at a cost different from A0 (cost per entered unit). **2.** Compute expected average = (Q0 x A0 + received base qty x new cost per base unit) / (Q0 + received base qty). **3.** Compare with the screen.	- GRN number issued; stock +10,000 KG. - Average cost equals your calculation (4 decimals). - GRN_POSTED audited.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

INV-02 Invalid receipts refused Role: yard

Steps	Expected result	Result
1. Try receiving quantity 0 and -5, and a negative cost.	- Refused (service or database constraint); no movement written.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

INV-03 Transfer leaves at frozen cost and is in transit Role: yard

Steps	Expected result	Result
1. Transfer 20 BAG of Product C from Yard 1 to Yard 2 (Transfer, then dispatch). **2.** Check both yards.	- TRF number; status IN_TRANSIT. - Yard 1 stock down at its average cost; Yard 2 not yet increased; value is held in transit.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

INV-04 Receive the transfer in full Role: yard (destination)

Steps	Expected result	Result
1. Receive the transfer at Yard 2 for the full sent quantity.	- Status RECEIVED; Yard 2 stock up by sent base quantity at the frozen cost.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

INV-05 Receive a shortfall and an over-receipt Role: yard

Steps	Expected result	Result
1. Create a second transfer of 20 BAG. **2.** Receive only 18 BAG and record the shortfall. **3.** Try to receive 25 BAG on a third transfer.	- A TransferDiscrepancy record shows the shortfall with value. - Receiving more than sent is refused (cumulative received cannot exceed sent).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

INV-06 Stock adjustment / count needs an explanation Role: yard

Steps	Expected result	Result
1. Adjust / count: enter a signed difference (-10 KG) with reason STOCKTAKE and explanation. **2.** Repeat leaving the explanation empty.	- First accepted, ADJ number issued, STOCK_ADJUST audited. - Empty explanation refused.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

INV-07 Negative stock raises an alert Role: yard, admin

Steps	Expected result	Result
1. Adjust a small-stock product below zero (authorised adjustment). **2.** Run `python manage.py check_alerts`.	- A negative-stock line is reported (exit code 2); sale is not rewritten.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

INV-08 Rebuild projections changes nothing Role: admin

Steps	Expected result	Result
1. Record all stock quantities and values. **2.** Run `python manage.py rebuild_balances`. **3.** Compare.	- Quantities and values are identical before and after (ledger equals projection).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

INV-09 Cashier and manager cannot post stock Role: cashier, manager

Steps	Expected result	Result
1. Look for Receive goods, Transfer and Adjust / count.	- Absent / 403 for both.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

INV-10 Ledger cannot be edited Role: admin

Steps	Expected result	Result
1. In Administration open stock movements. **2.** Try to edit or delete a movement.	- Read-only; edit and delete refused.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

11 Reports, exports and the accounting interface

Before you start: Run after sections 5-10 so there is data. Logged in as manager or accounts. Note the basis line printed on each report.

REP-01 Daily cash Role: accounts

Steps	Expected result	Result
1. Reports - daily cash for today.	- Verified and pending amounts separate, by method; totals agree with closed sessions and refunds.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

REP-02 Receivables aging reconciles to customer balance Role: accounts

Steps	Expected result	Result
1. Open `reports?tab=aging`. **2.** Pick Customer K and compare with the customer balance.	- Shows gross exposure, unapplied deposits and net per customer. - Total equals the sum of customer balances.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

REP-03 VAT report Role: accounts

Steps	Expected result	Result
1. Open the VAT report for the month. **2.** Check one invoice and one credit note.	- Tax equals 15% of net (rate is configuration). - Credit-note net and tax shown separately.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

REP-04 Operational margin versus invoiced sales Role: manager

Steps	Expected result	Result
1. Open margin by product and the invoiced-sales view. **2.** Compare for the partially delivered invoice (GATE-05, after first DO).	- Margin counts only delivered quantity x net price less returns, matched to frozen issue cost. - Invoiced sales is a separate, labelled view.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

REP-05 Excel exports are safe and correct Role: accounts

Steps	Expected result	Result
1. As admin create a product named =1+1 and a customer named @SUM(A1). **2.** Export the sales register and stock on hand to Excel and open them.	- Hostile names appear as literal text (no formula runs). - TOTAL cells remain real formulas.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

REP-06 Every report states its basis Role: accounts

Steps	Expected result	Result
1. Open each report tab.	- Each shows a basis statement; none claims to be a statutory return.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

REP-07 Accounting export with checksums Role: admin

Steps	Expected result	Result
1. Run `python manage.py accounting_export`. **2.** Open the CSV files and manifest. **3.** Verify each file with `sha256sum`.	- CSV for invoices, credit notes, receipts, refunds and stock value movements. - Checksums in the manifest match; reconciliation summary produced.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

REP-08 Reports refused to cashier and storekeeper Role: cashier, yard

Steps	Expected result	Result
1. Open the reports address.	- 403 or not shown.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

12 Audit trail and immutability

Before you start: Perform after earlier sections. Open the audit trail as admin.

AUD-01 Every controlled action is audited Role: admin

Steps	Expected result	Result
1. Search the audit trail for each type: PRICE_CHANGE, DISCOUNT, CREDIT_OVERRIDE, VOID_INVOICE, VOID_RECEIPT, CREDIT_NOTE, STOCK_ADJUST, GRN_POSTED, USER_ENROLLED.	- An entry exists for each action you performed with user, time and object.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

AUD-02 No secrets in the audit trail Role: admin

Steps	Expected result	Result
1. Search entries for the word password, token and PIN.	- Nothing sensitive is stored.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

AUD-03 Posted facts cannot be changed Role: admin

Steps	Expected result	Result
1. In Administration try to edit or delete: an invoice, a payment, a delivery order, an audit entry.	- All refused. Corrections exist only as void, credit note, adjustment or discrepancy.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

13 Security checks

Before you start: Environment A. For SEC-02 and SEC-05 use a terminal with curl.

SEC-01 Hostile text is displayed, not executed Role: admin, cashier

Steps	Expected result	Result
1. As admin create a product named `<img src=x onerror=alert(1)>`. **2.** As cashier search for it in the POS and add it to the cart.	- The text appears literally. No alert box. No console errors.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SEC-02 CSRF protection Role: tester

Steps	Expected result	Result
1. Using curl, send a POST to a form address with a valid session cookie but no CSRF token.	- Request refused (403).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SEC-03 Forbidden pages by direct address Role: all roles

Steps	Expected result	Result
1. For each role, type the address of a screen it should not use (reports, dispatch, adjust, administration).	- Always 403 / redirect, never data.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SEC-04 API without login Role: tester

Steps	Expected result	Result
1. Using curl, call an API address under `/api/` without credentials.	- Refused (401/403).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SEC-05 API with malformed numbers Role: tester

Steps	Expected result	Result
1. As an authenticated user send payment amounts: abc, NaN, Infinity, -1, 99999999999999999999, empty.	- Each returns 400 with a message; none returns 500.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

14 Cloud synchronisation, devices and recovery

Before you start: Environment B: one cloud node and one edge node (site MKL1, ideally also a second edge MKL2). Cloud has products, units, prices, customers, yards with owner site set to MKL1, and the node's internet link can be switched off.

SYNC-01 Register an edge device Role: admin (cloud)

Steps	Expected result	Result
1. On the cloud run `python manage.py register_device MKL1 "Mekelle main yard PC"`. **2.** Copy the token shown.	- Token is displayed once. It is stored only as a digest (running the command again does not show the old token).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-02 Initialise the edge from the cloud Role: admin (edge)

Steps	Expected result	Result
1. Edit .env: SERK_NODE_ROLE=edge, SERK_SITE_CODE=MKL1, cloud URL and token. **2.** Run `migrate` then `setup_roles`. **3.** Run `python manage.py sync_once`.	- Products, prices, customers, this site's users and the site's yard ledger arrive. - Never run seed_demo on an edge.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-03 Trade while offline Role: cashier, yard

Steps	Expected result	Result
1. Disconnect the edge from the internet. **2.** Sell, take payment, dispatch. **3.** Look at the sync indicator.	- Everything works. The sync pill shows changes queued.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-04 Reconnect and drain the queue Role: admin

Steps	Expected result	Result
1. Reconnect. Press Sync now (or `sync_once`). **2.** Look at the cloud for the documents.	- Queue empties; invoices, payments and movements appear on the cloud. - Document numbers are gap-free per site.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-05 Master data comes down Role: manager (cloud)

Steps	Expected result	Result
1. On the cloud change a price and add a customer. **2.** On the edge run a sync.	- The edge shows the new price and customer after one cycle.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-06 Foreign device cannot write another yard Role: release engineer

Steps	Expected result	Result
1. Run the automated cases in apps/sync tests (CloudIngestTests) or the multi-node rehearsal. **2.** Optionally push a movement for MKL1's yard with MKL2's token.	- Rejected; no stock row changed; a conflict alert is raised on the cloud.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-07 Revoke a lost device Role: admin (cloud)

Steps	Expected result	Result
1. Administration - Devices - revoke the MKL1 device. **2.** On the edge run a sync.	- The edge is refused (401) and shows the device as revoked; no more data accepted.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-08 Replace a node Role: admin

Steps	Expected result	Result
1. On the cloud run `register_device MKL1 --replace`. **2.** Try the old token. **3.** On a new machine: migrate, setup_roles, then `recover_node --epoch N`. **4.** If the old machine still works run export_unsent, then recover_unsent on the new one.	- Old token gets 401 on every endpoint. - New machine recovers documents, payments, ledger and numbering resumes after the last issued number. - Unsent events from the old machine go to a review queue and are not auto-posted.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-09 Offline authority lease (optional, long) Role: admin

Steps	Expected result	Result
1. Set a short SERK_LEASE_DAYS in a test edge (or advance the clock in the test host). **2.** Keep the edge offline past the lease. **3.** Try to confirm, dispatch, take payment and read data.	- Confirm, dispatch and payment are refused with a clear message; reading continues. - After reconnecting, trading resumes (lease renews on ping). - Users must log in again after OFFLINE_REAUTH_DAYS without contact.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-10 Cross-site transfer with shortfall Role: yard (both sites)

Steps	Expected result	Result
1. Transfer from an MKL1 yard to an MKL2 yard. **2.** Sync both nodes. **3.** Receive a smaller quantity at the destination.	- Only the destination site can receive; shortfall becomes a discrepancy with value; both sides agree after sync.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-11 Branch credit allowance from the cloud Role: admin (cloud), cashier

Steps	Expected result	Result
1. On the cloud allocate a BranchCreditAllowance for Customer K at MKL1 with an expiry. **2.** Sync. **3.** Confirm unpaid invoices up to and over the allowance. **4.** Let the allowance expire and retry.	- Credit allowed only up to allowance minus the branch's open exposure. - With no valid allowance, an unpaid sale is refused (must be paid).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SYNC-12 Enrol a branch user Role: admin		
Steps	**Expected result**	**Result**
1. On the edge run `python manage.py enroll_user cash1 --role CASHIER --site MKL1`. **2.** Log in as cash1 with the edge offline. **3.** Try to enrol an ADMIN identity.	- cash1 can log in offline; cloud learns the user at the next sync. - A branch cannot create privileged accounts (refused).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

15 Operations: health, backup and restore

Before you start: Environment C (PostgreSQL). Operators follow the Operations and Backup/Restore runbooks. Keep the backup key file safe and off the machine.

OPS-01 Health endpoint Role: operator

Steps	Expected result	Result
1. Run `curl -s https://HOST/healthz/`. **2.** Run `python manage.py check_alerts` and read the exit code.	- Status ok, exit code 0 when healthy; 503 / exit 2 when an alert fires.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

OPS-02 Alert conditions fire Role: operator

Steps	Expected result	Result
1. Stop sync and wait (or lower thresholds), create a negative stock line, leave a cash session open over 24 h (or test with a lowered threshold).	- check_alerts reports each condition: oldest unsent change, unapplied inbox, quarantined rows, negative stock, session open too long, stale backup.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

OPS-03 Create a backup Role: operator

Steps	Expected result	Result
1. Run `python manage.py backup`. **2.** Open the new .serkbak folder.	- Contains db.dump.gz.enc, manifest.json (SHA-256 and size per file), media archive if any. - UPLOADED marker only after the remote copy is confirmed. - Device tokens are not in the backup.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

OPS-04 Restore on a clean host and compare Role: operator, accountant

Steps	Expected result	Result
1. Record a snapshot: stock quantity and value per yard, open partial obligations, receivables, audit row count, last issued numbers, unsent outbox count. **2.** On a clean host run `manage.py restore ./NAME.serkbak --key-file backup.key`. **3.** Run `preflight` and `check_alerts`. **4.** Take the same snapshot and compare.	- Snapshots are identical. - Projection integrity issues: 0. - Time taken is recorded against the 4 h RTO target (D17).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

OPS-05 Wrong key or corrupt backup Role: operator

Steps	Expected result	Result
1. Restore with a wrong key file. **2.** Edit one byte in a copy of the backup and restore it. **3.** Run backup in production with the key file missing.	- Each fails with a clear message and nothing on the target is changed. - Backup refuses to run without a key.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

OPS-06 Retention and offsite queue Role: operator

Steps	Expected result	Result
1. Disconnect the remote store and run a backup. **2.** Reconnect and run the next backup. **3.** Create more backups than SERK_BACKUP_KEEP.	- First run exits non-zero and queues the upload; next run uploads all queued backups. - Old generations are pruned only if their remote copy is confirmed.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

OPS-07 Nightly job Role: operator

Steps	Expected result	Result
1. Run the nightly timer manually: systemctl start serk-nightly.service. **2.** Read journalctl -u serk-nightly.	- Proformas expired, projections rebuilt, backup taken, accounting export written, alerts checked.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

OPS-08 Power loss on a branch PC Role: operator

Steps	Expected result	Result
1. During an active sale kill the web process and database processes (or pull the plug on the rehearsal PC). **2.** Restart. **3.** Run preflight and rebuild_balances.	- No half-written documents; ledger equals projection.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

OPS-09 Upgrade rehearsal Role: operator

Steps	Expected result	Result
1. Follow the upgrade procedure: backup, stop, preflight, migrate as owner, setup_roles, rebuild_balances, re-grant, preflight, check_alerts, start. **2.** Compare customer balances and stock before and after.	- No blocking preflight finding; figures identical before and after; /healthz/ ok.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

OPS-10 Monthly restore test is performed Role: operator

Steps	Expected result	Result
1. Restore last night's offsite backup on a standby machine (database serk_restore_test). **2.** Run the comparison and write it in the operations log.	- Result filed with date, backup name, figures and operator.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

16 Deployment and production-profile checks

Before you start: Environment C. Fill the two env files with test values. Do this on the actual branch hardware for the NOT RUN gates.

DEP-01 Configuration fails closed Role: release engineer

Steps	Expected result	Result
1. With a production profile run `python manage.py check` once for each bad setting: **2.** (a) DEBUG on, (b) dev secret key, (c) wildcard allowed hosts, (d) missing CSRF origins, (e) SQLite under branch/cloud, (f) non-https cloud URL.	- Each case is refused with a clear message (the runbook counts five refusals but lists six conditions - test all six and note the difference). - The correct config passes `check --deploy --fail-level WARNING`.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

DEP-02 Runtime database role is least-privilege Role: release engineer

Steps	Expected result	Result
1. Connect as serk_app and try TRUNCATE on a ledger table, dropping a trigger and altering a constraint. **2.** Update a posted payment with SQL.	- All refused. Normal application activity still works.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

DEP-03 TLS and trust Role: release engineer

Steps	Expected result	Result
1. Open the branch address (mkl1.serk.local) from a terminal and from a tablet with the internal CA installed. **2.** Try a tablet without the CA.	- Valid padlock with the CA; certificate warning without it. HTTP redirects to HTTPS; secure cookies and HSTS present.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

DEP-04 Service control Role: release engineer

Steps	Expected result	Result
1. systemctl reload serk-materials while someone is selling. **2.** Kill the web process; watch it restart. **3.** Stop and start sync.	- Reload causes no failed requests. Services restart after failure (bounded to 5 per 300 s).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

DEP-05 Container image and supply chain Role: release engineer

Steps	Expected result	Result
1. Verify the release ZIP with `sha256sum -c SHA256SUMS`. **2.** Pin the base-image digest at build time and scan the image (Trivy or Grype).	- Hashes match; scan report has no unaccepted critical findings.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

DEP-06 Alert delivery on the target host Role: operator		
Steps	**Expected result**	**Result**
1. Make serk-alerts.service fail on purpose.	- Mail or SMS hook delivers the alert to a named person.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

DEP-07 Load on target hardware Role: release engineer		
Steps	**Expected result**	**Result**
1. Run 8 terminals creating and confirming invoices for 10 minutes with sync and nightly running.	- No errors or deadlocks; create+confirm p95 comfortably under 1 s (the reference run was 0.67 s on 2 vCPU - re-measure).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

17 Automated test suite and evidence

The automated suite is run by `tools/verify.sh` (needs PostgreSQL reachable through PGHOST, PGPORT, PGUSER and Chromium via Playwright). Every layer writes its command, backend, git hash and exit code to evidence/run_STAMP/.

Layer	Backend	Expected result
Static checks (check, deploy check, makemigrations --check, compileall, lock dry-run)	-	6 of 6 ok
Unit, domain, DB guards, autocommit (manage.py test apps)	SQLite	92 run, 85 ok, 7 skipped (PG races), 0 failures
Same suite	PostgreSQL 16	92 run, 92 ok, 0 skipped
Independent 38 safety checks (successor module)	SQLite and PostgreSQL	38 of 38 on both
Randomised conservation (40-step sequences)	PostgreSQL	all seeds ok
Native races (7)	PostgreSQL	7 of 7
Multi-node (cloud + MKL1 + MKL2)	PostgreSQL	16 of 16
Backup/restore drill	PostgreSQL	12 of 12
Real-browser journeys	Chromium 141	23 of 23, no unexpected console errors
Deployment rehearsal	PostgreSQL	20 of 20
Dependency scan and SBOM	pip-audit, CycloneDX	0 known vulnerabilities

The unmodified *original* 38-check module is expected to show 19 pass and 19 fail or error on the candidate. This is informational: all failures come from deliberate interface changes. The *successor* module is the active control.

Manual cross-check

- Run the suite on the actual release ZIP and compare the git hash printed in the evidence with the candidate hash in RELEASE_READINESS.
- File the evidence folder with the signed manual results as the acceptance record.

18 Scope confirmation and owner acceptance checks

These tests confirm that the system honestly **does not** contain features that the owners must not assume exist (D14, D15, D16).

SCOPE-01 Excluded features are really absent Role: product owner

Steps	Expected result	Result
1. Look in the menus and reports for: general ledger or journals, posting periods, purchase orders and payables, bank reconciliation, weighbridge tickets, lot or condition tracking, offcut inventory, gate-operator role, fiscal receipt printing, Amharic or Tigrinya screens, Ethiopian calendar, a second currency, live Telebirr/CBE/bank verification.	- None of these exists. Everything is English and ETB only. - Owner acknowledges the exclusions in writing (D14, D15).	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SCOPE-02 VAT rate is configuration only Role: finance / tax advisor

Steps	Expected result	Result
1. Confirm the rate in settings (SERK_VAT_RATE = 15%). **2.** Compare invoice contents and TIN with the company registration and current ERCA/MoR rules.	- Tax advisor confirms rate, TIN, invoice content and fiscal-receipt obligations before cutover (D16). No compliance is claimed by the software.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

SCOPE-03 Owner accepts policies D01-D17 Role: owners

Steps	Expected result	Result
1. Walk the owners through the traceability table in Appendix D and show the real behaviour for each. **2.** Each owner signs the accepted column in DECISION_REGISTER.md.	- All 17 policies signed or a change is requested and logged as a defect or change request.	[ ] Pass [ ] Fail [ ] Blocked Tester / date: Notes / defect no.:

19 End-to-end day scenario (user acceptance)

Before you start: Fresh demo database. Write the start figures in Appendix C: S0 = stock of Product C (KG), A0 = average cost, K0 = Customer K balance. Use one price list throughout. All expected figures are formulas of your start figures.

#	Who	Action	Expected result / figure to check
1	cashier	Open cash session with float 1,000.00	Session open
2	cashier	Walk-in sells 10 BAG of C. Confirm, collect CASH in full	Receipt RCT, invoice settled; ATP down 500 KG
3	yard	Dispatch the walk-in invoice	DO issued; stock = S0 - 500; cost at A0
4	cashier	Customer K buys 100 BAG on credit (inside allowance). Confirm	Balance K0 + invoice total; available credit reduced
5	yard	Dispatch 40 BAG now	Status PARTIAL, remaining 60; stock = S0 - 500 - 2,000
6	cashier	Contractor pays a different invoice by TELEBIRR with a reference	PENDING; goods cannot leave yet
7	accounts	Verify the Telebirr payment	VERIFIED; invoice settled; yard may dispatch
8	yard	Receive 200 BAG at a different cost	Stock +10,000 KG; new average = (Q x A + 10,000 x new cost) / (Q + 10,000)
9	yard	Dispatch the remaining 60 BAG to Customer K	FULFILLED; stock down a further 3,000 KG
10	manager	Customer K returns 5 BAG (RESTOCK). Credit note. Refund the cash portion	CRN issued; stock +250 KG at original issue cost; refund recorded against session
11	cashier	Close session: counted = float + cash receipts - cash refunds	Variance 0.00, closes without approval
12	accounts	Reports: daily cash, aging, VAT, margin	Cash agrees with session; aging total = customer balances; VAT 15% of net; margin on delivered quantity only
13	operator	Run nightly steps and a backup; restore on a second machine	Snapshots equal (OPS-04)
14	yard	Final stock of Product C	S0 - 500 - 2,000 - 3,000 + 10,000 + 250 KG exactly

Result of the end-to-end scenario: [ ] Pass [ ] Fail Date: _____ Accountant: _____
Product owner: _____

20 Open gates and pilot checks (from the release readiness record)

These items are marked NOT RUN or BLOCKED in the release notes. They must be closed before any live yard is cut over.

#	Gate	How to close	Owner	Done
1	Policies D01-D17 accepted	SCOPE-03, Appendix D	Product, Finance, Operations, Security	[ ]
2	Ethiopian VAT / fiscal obligations verified	SCOPE-02	Finance / tax advisor	[ ]
3	Launch scope confirmed (POS + external accounting)	SCOPE-01	Product	[ ]
4	Live payment-provider verification	Not available; manual verification (PAY-05) is what ships	Finance / provider	[ ]
5	Deployment rehearsal on actual branch hardware and LAN	Section 16 on target PCs, UPS, internal CA on tablets	Operations	[ ]
6	Offsite restore from the real remote location	OPS-04, OPS-06 using the real S3/remote	Operations	[ ]
7	Docker base-image pin and scan	DEP-05	Release engineer	[ ]
8	Pilot 2-4 weeks beside the paper process	Daily compare of counts, cash and receivables	Product, Accountant	[ ]
9	Alert delivery on target host	DEP-06	Operations	[ ]

Cutover checklist (first live day)

- Physical stock count per yard entered as an OPENING adjustment with unit cost; accountant signs the count sheet.
- Opening receivables entered per customer and reconciled: the total of reports?tab=aging must equal the accountant's ledger figure.
- Branch credit allowances allocated on the cloud; walk-in and cash policy confirmed with the owner.
- First backup taken and restored on a second machine.
- Tax advisor confirmation of VAT rate, TIN, invoice content and fiscal receipt obligations.

Pilot daily comparison sheet

Date	Cash per system	Cash counted	Stock count variance	Receivables vs paper	Issues

Appendix A Defect log

No.	Test ID	Severity	Description and steps to reproduce	Found by / date	Status
D-01					
D-02					
D-03					
D-04					
D-05					
D-06					
D-07					
D-08					
D-09					
D-10					
D-11					
D-12					
D-13					
D-14					

Appendix B Test summary and sign-off

Section	Tests	Pass	Fail	Blocked
3 Login and access	8			
4 Catalog and prices	5			
5 Cash sessions	5			
6 Sales	14			
7 Payments	10			
8 Gate and dispatch	10			
9 Returns and refunds	6			
10 Inventory	10			
11 Reports	8			
12 Audit	3			
13 Security	5			
14 Sync	12			
15 Operations	10			
16 Deployment	7			
18 Scope and acceptance	3			
19 End-to-end scenario	1			

Release recommendation (tick one): [ ] GO for pilot [ ] NO-GO [ ] GO with listed conditions

Role	Name	Signature	Date
Lead tester			
Accountant / Finance			
Operations			
Security			
Product owner			

Appendix C Test data sheet

Fill in before testing and update as you go.

Item	Value	Item	Value
Product C name		Product R name	
S0 stock (KG) Yard 1		S0 stock Yard 2	
A0 average cost (per KG)		Product C BAG price P0	
Customer K name		K credit limit	
K branch allowance		K0 opening balance	
Customer Z (limit 0)		Cash float used	

Doc type	Numbers created (list)	Related test
PRO (proforma)		
INV (invoice)		
DO (delivery)		
RCT (receipt)		
GRN		
TRF (transfer)		
ADJ (adjustment)		
CRN (credit note)		

Appendix D Decision traceability (D01-D17)

ID	Policy (short)	Verified by tests	Owner signed
D01	Confirmation reserves stock; backorder needs capability	SAL-02, SAL-06, SAL-07, SAL-08, SAL-12	[ ]
D02	Walk-in is a cash sale	SAL-11, GATE-01, GATE-02, GATE-03	[ ]
D03	Credit is a branch allowance	SAL-09, SAL-10, SYNC-11	[ ]
D04	Non-cash is pending until a second user verifies	PAY-02 to PAY-06, GATE-03	[ ]
D05	Cash sessions and explicit refunds	CASH-01 to CASH-05, RET-05	[ ]
D06	Weighted average valuation, original cost on returns	INV-01, INV-08, RET-01, RET-03, GATE-10	[ ]
D07	Transfers freeze at dispatch; discrepancy documents	INV-03 to INV-05, SYNC-10	[ ]
D08	Report bases	REP-01 to REP-06	[ ]
D09	One authenticated stock authority per yard	SYNC-01, SYNC-06, SYNC-07	[ ]
D10	Offline authority lease	SYNC-09	[ ]
D11	Node replacement and fencing	SYNC-08	[ ]
D12	Identity administration is not synced	SYNC-12, AUTH tests	[ ]
D13	PostgreSQL required for production	DEP-01	[ ]
D14	Launch scope: POS + external accounting	REP-07, SCOPE-01	[ ]
D15	Construction-specific measures not built	SCOPE-01	[ ]
D16	Ethiopian tax is configuration only	REP-03, SCOPE-02	[ ]
D17	RPO 24 h / RTO 4 h	OPS-03, OPS-04, OPS-10	[ ]

Appendix E Role and capability reference

Capability	Admin	Manager	Cashier	Storekeeper	Accountant
Sell / discount limit	Yes 100%	Yes 15%	Yes 3%	-	-
Override credit, declare backorder	Yes	Yes	-	-	-
Collect payment	Yes	Yes	Yes	-	Yes
Verify non-cash payment (not own)	Yes	Yes	-	-	Yes
Void / credit note	Yes	Yes	-	-	Yes
Dispatch	Yes	-	-	Yes	-
Receive stock, transfer, adjust / count	Yes	-	-	Yes	-
View cost and margin	Yes	Yes	-	-	Yes
Manage prices	Yes	Yes	-	-	-
Reports	Yes	Yes	-	-	Yes
Rebuild projections / data repair	Yes	-	-	-	-

Use this table as the expected result for any role-based test. If the software and this table disagree, log a defect and check the DECISION_REGISTER and ARCHITECTURE section 2 for the authoritative rule.